

CIS270A Project 4 and Project 5 Requirements

1. Review the following items before starting:
 - a. Chapter 12 Selected Topics Powerpoint within the “First RPG Course Content” unit.
 - b. “Interactive Query Video” within the “First RPG Course Content” unit.
 - c. This video provide more demonstrations of using RDi Screen Designer. It is produced by a highly regarded trainer and I recommend watching it.
<https://youtu.be/HY9327WunR8>
2. To get the full benefit of this assignment, start from “scratch” in RDi Screen Designer. Also, it is good to become skilled at working with both your screen definition DDS and the RPG program that works with it in RDi. This video demonstrates the steps of starting from scratch, adding a “record” to an existing definition, and how to move in between your DDS/Screen and RPG code, as well as a tip on what to do if your RDi “times out.”
Clip 1 (starting from scratch and what to do if RDi times out)
<https://youtu.be/JFknmrJciVM>
 - b. Clip 2 (using ERRMSG and working with the source code view)
<https://youtu.be/S0cN421zTxg>
 - c. Clip 3 (adding screen records to your definition, toggling between RPG and DDS code; note that the RPG code in this clip is fixed form, however the concept is the same)
<https://youtu.be/ikdrH5JnIKQ>
3. Your next RPG program will be an interactive program, therefore you need to design the screens before writing the RPG program. It is important to understand what your RPG program will need to do before designing the screens that will interact with it.
 - a. A physical file called POINTSP has been defined as follows with the RPGIV_5THS library:

A	R POINTSREC		
A	STUNO	9S 0	TEXT('STUDENT NUMBER')
A	ASSIGNDESC	20A	TEXT('ASSIGNMENT DESCRIPTION')
A	PTSRECEIVE	3S 0	TEXT('POINTS RECEIVED')
A	PTSPOSSIBL	3S 0	TEXT('POINTS POSSIBLE')

Notes about this file: this is a sequential file (no key) that an instructor uses to record various scores for assignments. Notice that the student’s name is NOT present in this file. The student’s name is found in the WUSTDP file.

4. View the DDS for WUSTDP, which is in QDDSSRCW inside the RPGIV_5THS library. Take note of the key to the file and the field names for the student’s first and last names.
5. *You will be copying the POINTSP file to MYPOINTSP in your own library* (see Project 2 sheet for method)
6. After copying the file, your MYPOINTSP file will be currently empty. It has been defined and compiled, but does not contain any records. Your RPG program will write records to this disk file. These are the behaviors your program should exhibit:

- a. The first screen your user sees will ask them to enter the student number. When a number is entered, your program will randomly (chain) retrieve the student's name from WUSTDP. If there is no such record, an error message should display.
- b. When a successful chain has occurred, the next screen your user sees displays the student's name (as output only) and asks them to enter the homework description, points received, and points possible.
 - i. The points received cannot exceed the points possible, so if this occurs, an error message should display. Recall that an indicator can be "turned on" and then that indicator can be used to condition your ERRMSG.
 - ii. The points possible cannot be zero or negative; so if this occurs an error message should display.
 - iii. See "Data Validation in an Interactive Application" in the Blackboard unit.
- c. When your user has entered satisfactory data, this data should be written to the MYPOINTSP data file. See Chapter 9 regarding how to properly define your MYPOINTSP file so that records can be added, and how to add records to a disk file.
- d. Next, the last screen should show the student's percentage for this assignment, along with a text field that either says "Student passed" or "Student failed" (based on 60 percent or more being a passing percentage). Note that these items are for display purposes only and will not be written to the disk file, and should be output only.
- e. When the user presses enter on the last screen, the program should display the first screen for continued use of the application until the user decides to exit.

PROJECT 4 REQUIREMENTS:

Use RDi Screen Designer to design the three screens described above. Take screen captures of each screen in design mode, proving that you used the Screen Designer tool to create the screens. Place the three screen captures into a word processing document and upload the document in Blackboard, using the "Project 4 Upload Link."

In addition to the items needed to exhibit the behaviors listed above, your screens should contain the following elements:

1. Your Name
2. A name for your application
3. The current date
4. F3 to exit the application
5. Use of color and any other attributes that you decide upon, avoid light colors.

PROJECT 5 REQUIREMENTS:

Write the RPG program that interacts with these screens as described above. *Be sure to use MYPOINTSP as your disk file for the program, by copying POINTSP into MYPOINTSP in your library.* Test your program thoroughly, including the behaviors described above and disk file content. ***Keep in mind that your users are unpredictable, test thoroughly!***

- Does F3 work on every screen?
- If user presses F3 while on second screen, data should NOT be written to the file
- Are the appropriate error messages displaying?
- Test passing and failing, be sure to test 100 points out of 100 points as well
- Only write data to the file if the data is valid

RUNQRY is an easy way to view disk file content. Do your housekeeping as described in earlier projects and email me when you are ready to have your program evaluated.